

CERVIA, Italy

WMO: 161480

Lat: 44.224N

Lon: 12.307E

Elev: 18

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.1	28.1	19.2	14.5	33.8	22.8	17.2	33.6	23.9	40.6	21.0	40.2	6.4	270	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	91.5	75.3	89.4	75.0	86.9	74.4	79.8	86.1	78.1	85.0	76.5	83.8	10.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.4	147.5	83.7	75.8	134.9	82.0	73.8	126.2	80.6	43.4	86.9	41.5	85.6	39.9	84.5	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	17.7	15.0	20.5	96.3	4.1	2.7	17.5	98.2	15.0	99.8	12.7	101.3	9.7	103.3	
			DB												
			WB	20.2	83.3	4.0	1.8	17.3	84.6	15.0	85.7	12.8	86.7	9.9	88.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	39.1	41.6	48.2	55.1	63.5	71.1	75.5	75.2	67.2	59.2	49.6	40.6
	DBStd	13.86	4.90	5.40	5.64	4.92	4.87	5.05	4.17	4.52	5.12	5.46	6.37	5.63
	HDD50	1069	338	237	99	13	0	0	0	0	4	83	295	
	HDD65	3816	801	655	520	297	88	9	0	1	34	193	462	756
	CDD50	3711	1	2	43	167	417	632	789	780	516	289	71	4
	CDD65	984	0	0	0	1	40	191	324	315	100	12	1	0
	CDH74	9607	0	0	0	14	358	1853	3460	3182	689	50	1	0
	CDH80	3342	0	0	0	2	55	569	1338	1235	141	2	0	0
Wind	WSAvg	6.2	6.0	6.3	6.7	6.4	6.0	6.1	6.5	6.2	6.1	5.6	6.1	6.2
Precipitation	PrecAvg	26.8	1.5	1.7	2.2	2.4	2.0	1.9	1.4	1.9	3.0	3.1	3.0	2.2
	PrecMax	37.7	4.3	5.3	5.0	4.5	5.7	5.2	4.3	3.9	6.3	6.7	6.2	6.8
	PrecMin	18.5	0.1	0.1	0.4	0.9	0.5	0.3	0.0	0.1	1.0	0.4	0.5	0.3
	PrecStd	5.6	1.1	1.2	1.3	1.0	1.3	1.1	0.9	1.2	1.5	1.8	1.6	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	61.9	69.5	75.3	85.7	91.5	93.6	95.1	87.7	78.9	70.0	60.5
		MCWB	51.8	53.1	56.6	63.6	69.1	74.2	75.2	75.8	73.9	68.8	61.2	54.5
	2%	DB	53.6	57.4	65.8	71.6	80.8	88.5	91.4	91.8	84.1	75.0	64.8	55.3
		MCWB	48.1	49.8	55.5	61.4	67.3	74.5	75.4	75.6	73.2	66.7	59.5	51.1
	5%	DB	49.9	53.9	62.5	68.4	78.3	86.0	89.2	89.5	80.7	71.6	62.4	51.6
		MCWB	46.5	48.5	54.0	59.5	67.1	73.6	74.8	75.4	70.8	65.6	58.9	48.4
	10%	DB	46.7	51.4	59.2	66.1	75.1	83.0	86.4	86.4	77.4	68.3	59.2	48.6
		MCWB	44.2	47.0	52.5	58.4	65.0	71.7	74.1	74.9	68.9	63.5	56.4	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	54.4	59.6	66.6	73.2	80.5	82.4	82.3	78.7	71.3	63.7	55.5
		MCDB	56.5	59.7	65.4	72.1	79.0	86.3	87.6	86.9	83.3	75.1	66.3	58.9
	2%	WB	49.5	51.5	57.0	63.7	70.6	77.8	79.7	79.8	75.5	68.7	61.0	51.7
		MCDB	52.2	55.4	62.6	68.9	76.9	84.5	86.1	86.4	81.0	72.3	64.1	54.7
	5%	WB	46.8	49.7	55.3	61.3	68.6	75.6	77.9	78.1	72.8	66.5	59.1	49.3
		MCDB	49.1	53.1	60.7	66.1	75.0	82.5	84.8	85.4	78.3	70.2	61.7	51.1
	10%	WB	44.5	47.6	53.3	59.4	66.7	73.6	76.1	76.4	70.5	64.4	56.9	46.7
		MCDB	46.1	50.6	58.2	64.2	73.0	80.6	83.7	84.0	75.8	67.9	59.4	48.4
Mean Daily Temperature Range	5% DB	MDBR	12.3	15.7	17.6	17.7	18.6	19.5	20.0	19.8	18.1	15.2	12.2	11.7
		MCDBR	17.9	20.9	22.8	22.3	23.1	22.6	22.7	23.0	21.0	18.9	15.2	15.8
		MCWBR	13.8	14.9	14.7	14.2	13.2	12.2	12.3	12.8	12.3	12.5	11.5	12.3
	5% WB	MCDBR	15.4	18.9	20.3	19.8	20.3	20.6	20.4	21.0	18.7	17.2	13.4	13.9
		MCWBR	12.7	14.4	14.4	14.1	13.4	13.0	12.8	13.3	12.6	12.8	11.0	11.7
Clear-Sky Solar Irradiance	taub	0.355	0.375	0.408	0.442	0.441	0.449	0.441	0.434	0.422	0.411	0.379	0.351	
	taud	2.403	2.321	2.237	2.169	2.221	2.244	2.272	2.308	2.317	2.324	2.346	2.431	
	Ebn at Noon	239	254	260	261	265	262	263	260	253	238	225	228	
	Edh at Noon	25	31	39	45	44	43	41	39	36	31	26	22	
All-Sky Solar Radiation	RadAvg	426	698	1107	1473	1837	2086	2130	1824	1329	835	479	379	
	RadStd	64	104	140	140	132	145	108	97	115	91	89	56	49

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+3.20	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	-162	+153	+83

Nomenclature: See separate page